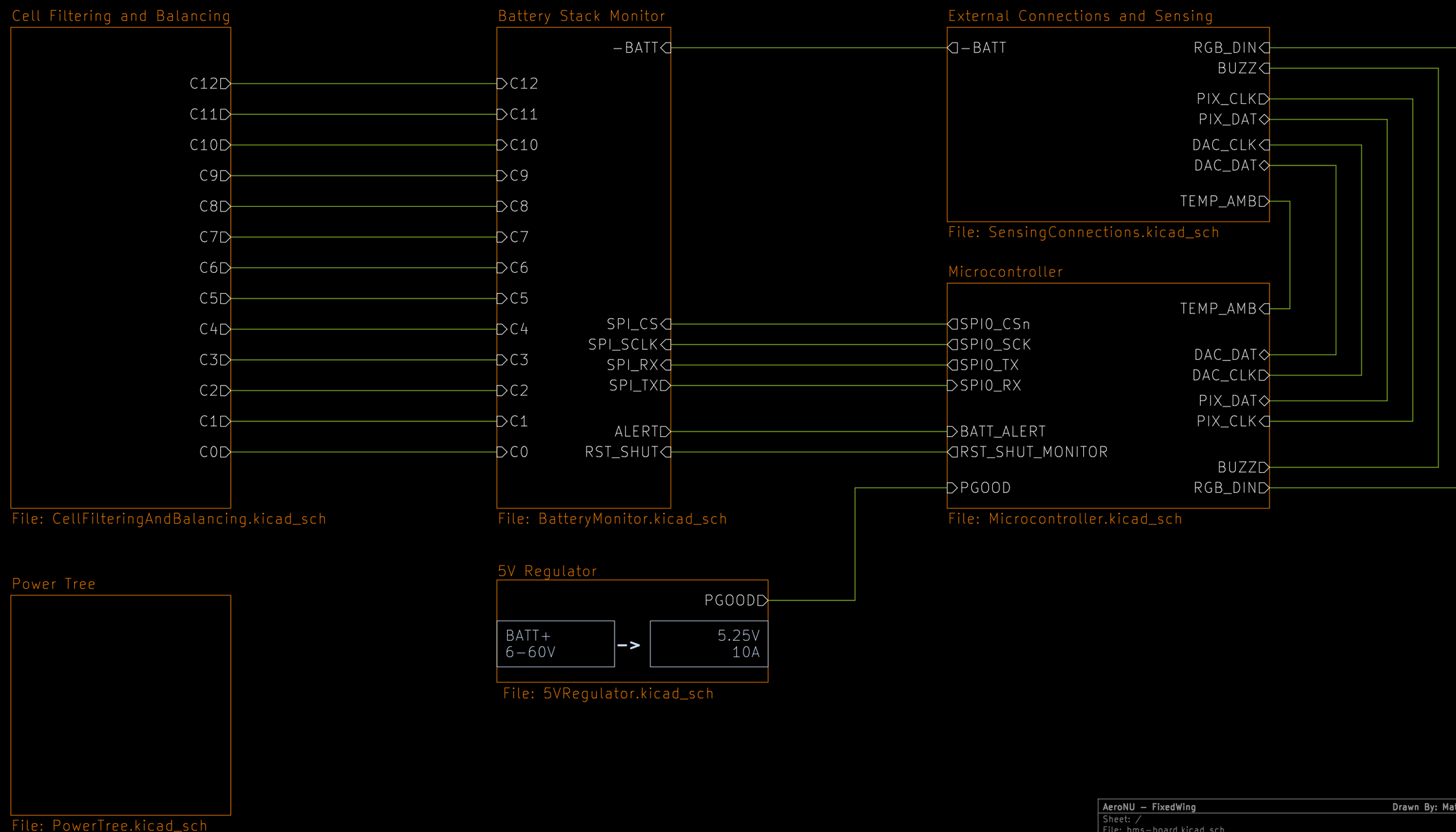
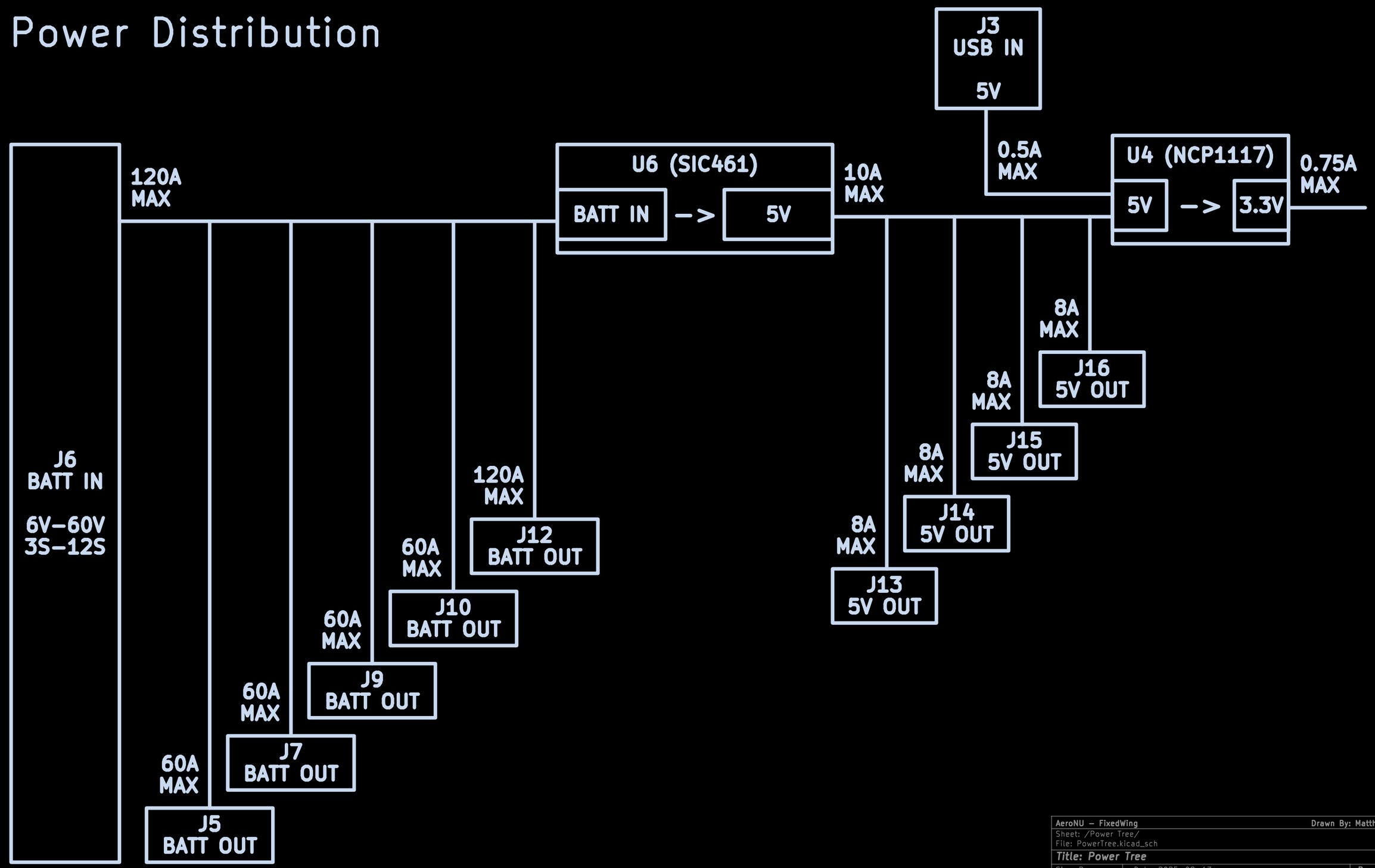
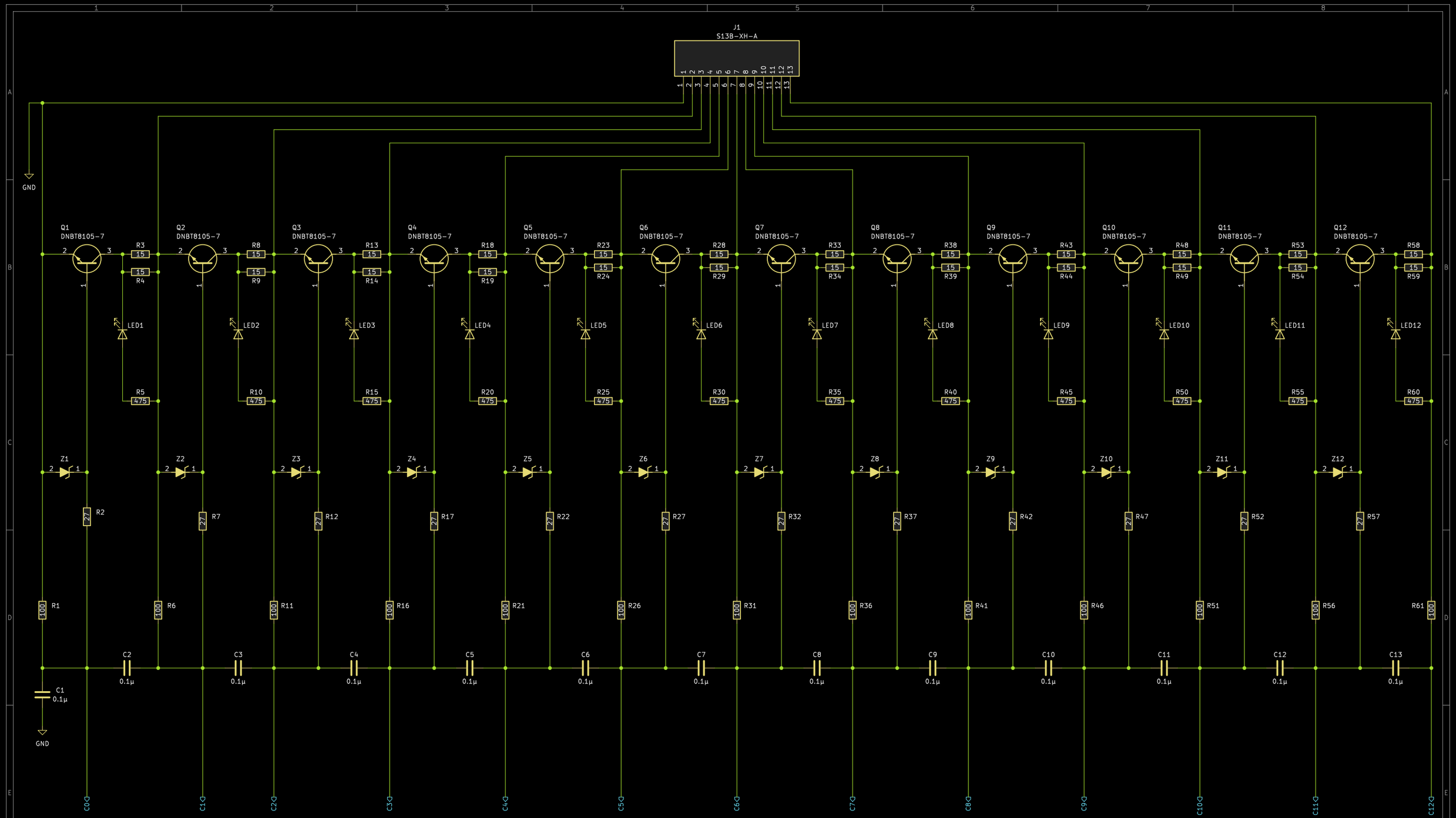


AeroNU – FixedWing	Drawn By: Matthew Friedman
Sheet: /	
File: bms−board.kicad_sch	
Title: Top−Level Interconnect	
Size: B / Date: 2025−08−13	Rev: A
KiCad E.D.A. kicad 7.0.10	Id: 1/7

Power Distribution



AeroNU – FixedWing	Drawn By: Matthew Friedman
Sheet: /Power Tree/	
File: PowerTree.kicad_sch	
Title: Power Tree	
Size: B / Date: 2025-08-13	Rev: A
KiCad E.D.A. kicad 7.0.10	Id: 2/7

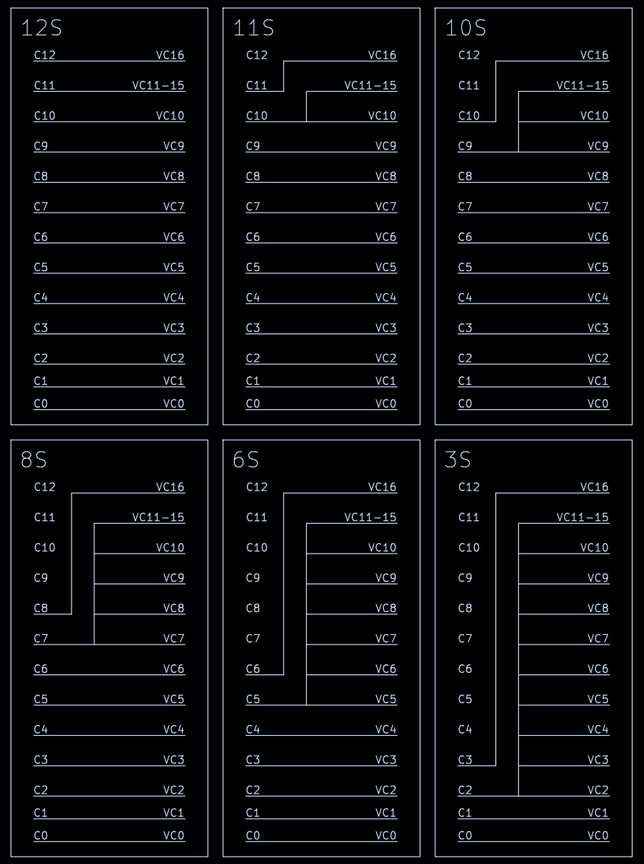


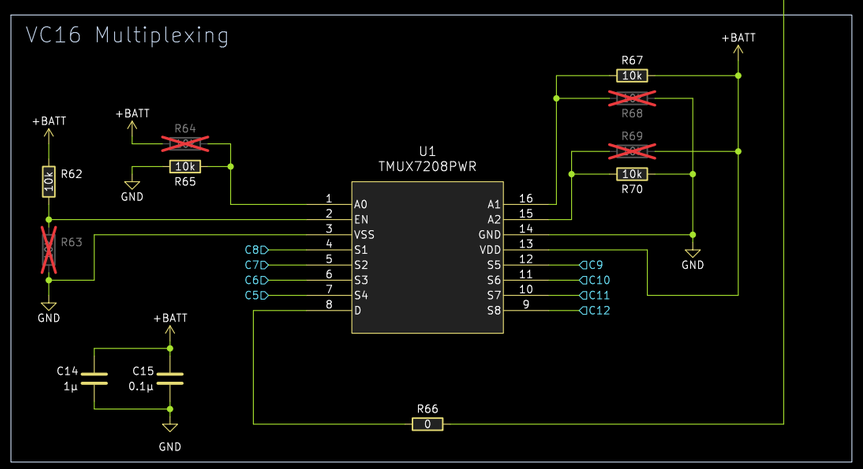
NOTE:
Design based on section 2.4 (External Cell-Balancing Circuit Design Using BJTs) of the Cell Balancing With BQ769x2 Battery Monitors guide by TI, available at http://www.ti.com/lit/pdf/sluaa81

AeroNU – FixedWing	Drawn By: Matthew Friedman
Sheet: /Cell Filtering and Balancing/ File: CellFilteringAndBalancing.kicad_sch	
Title: Cell Filtering and Balancing	
Size: B — Date: 2025-08-13	Rev: A
KiCad E.D.A. kicad 7.0.10	Id: 3/7

CONFIGURATION:

- The 0Ω jumpers along with the multiplexer allow for board-specific configurations for varying cell counts from 3S through 12S, as shown in the drawings below. The current schematic is DNP'd for a 6S configuration.
- The jumpers allow users to tie unused pins together correctly, while the multiplexer ensures that the topmost VC pin (VC16) is connected to the correct cell input.
- It is highly important that users who are populating the jumpers and pull-up/down resistors on the select lines for the multiplexer understand the underlying circuitry for their desired cell count, as there exists a real risk of causing a short.

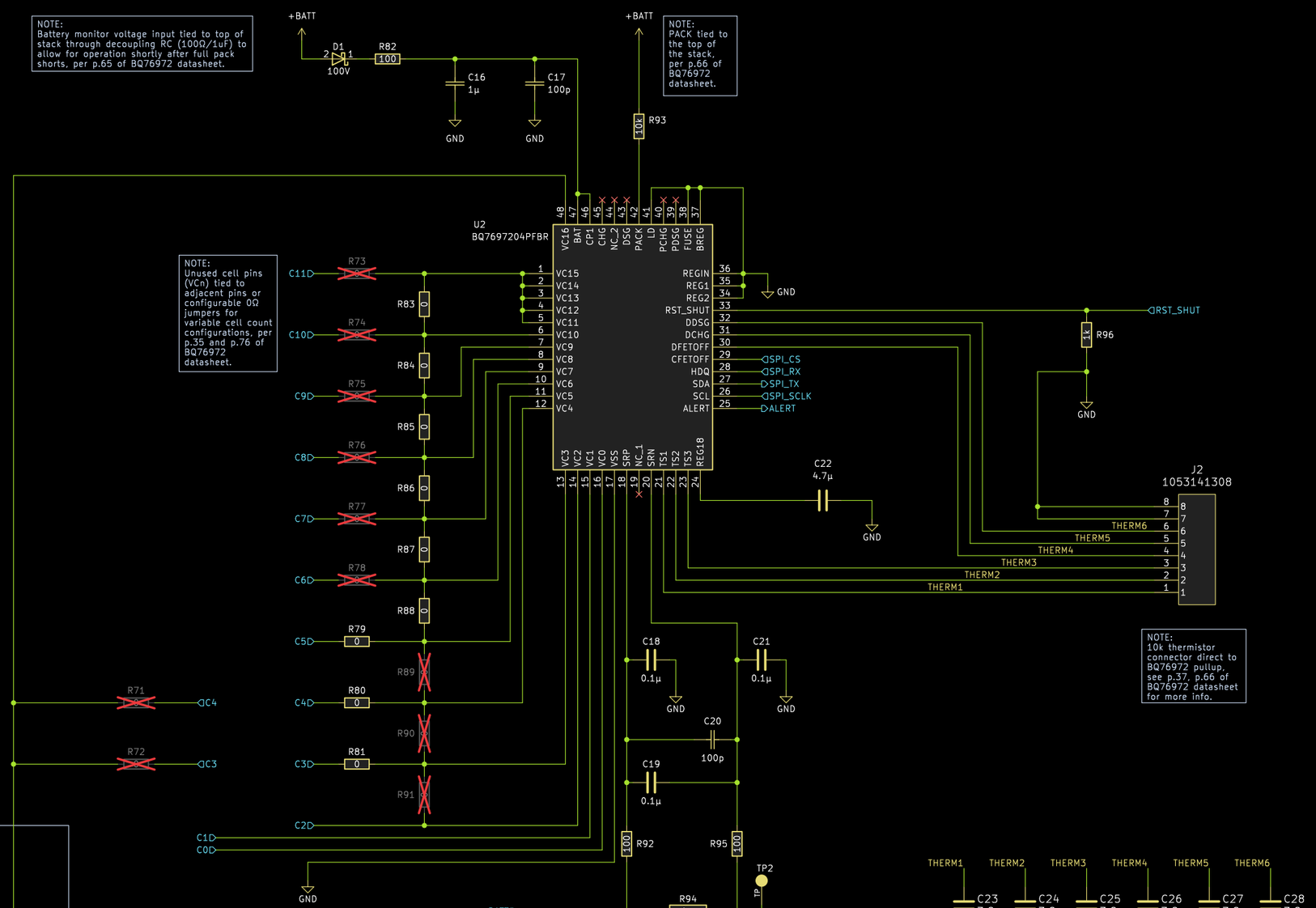




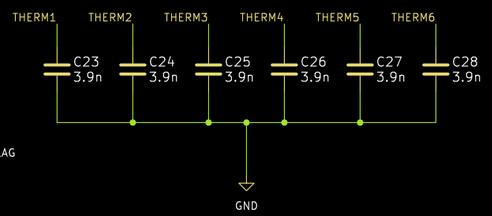
NOTE:
Battery monitor voltage input tied to top of stack through decoupling RC (100Ω/1uF) to allow for operation shortly after full pack shorts, per p.65 of BQ76972 datasheet.

NOTE:
PACK tied to the top of the stack, per p.66 of BQ76972 datasheet.

NOTE:
Unused cell pins (VCn) tied to adjacent pins or configurable 0Ω jumpers for variable cell count configurations, per p.35 and p.76 of BQ76972 datasheet.



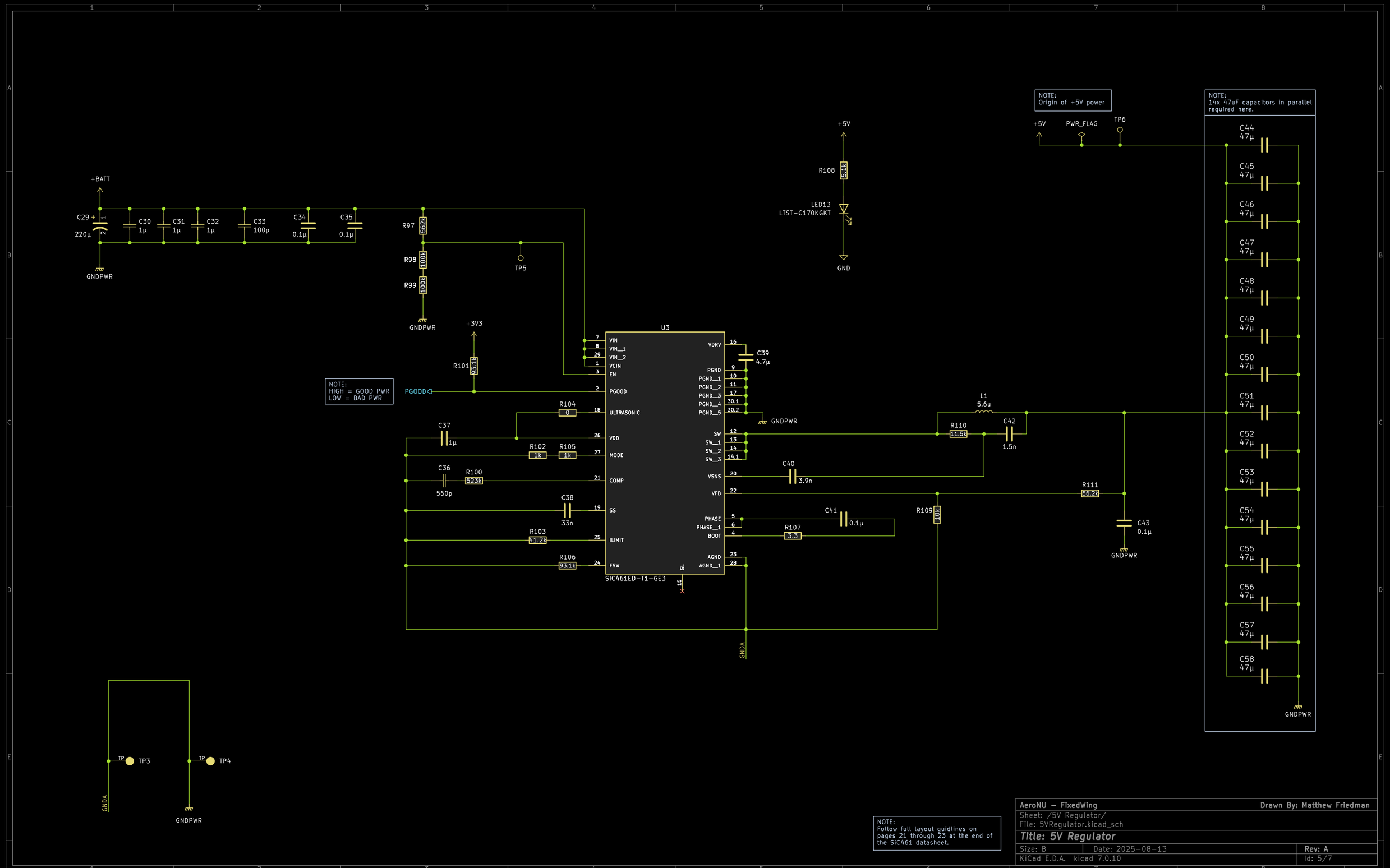
NOTE:
10k thermistor connector direct to BQ76972 pullup, see p.37, p.66 of BQ76972 datasheet for more info.



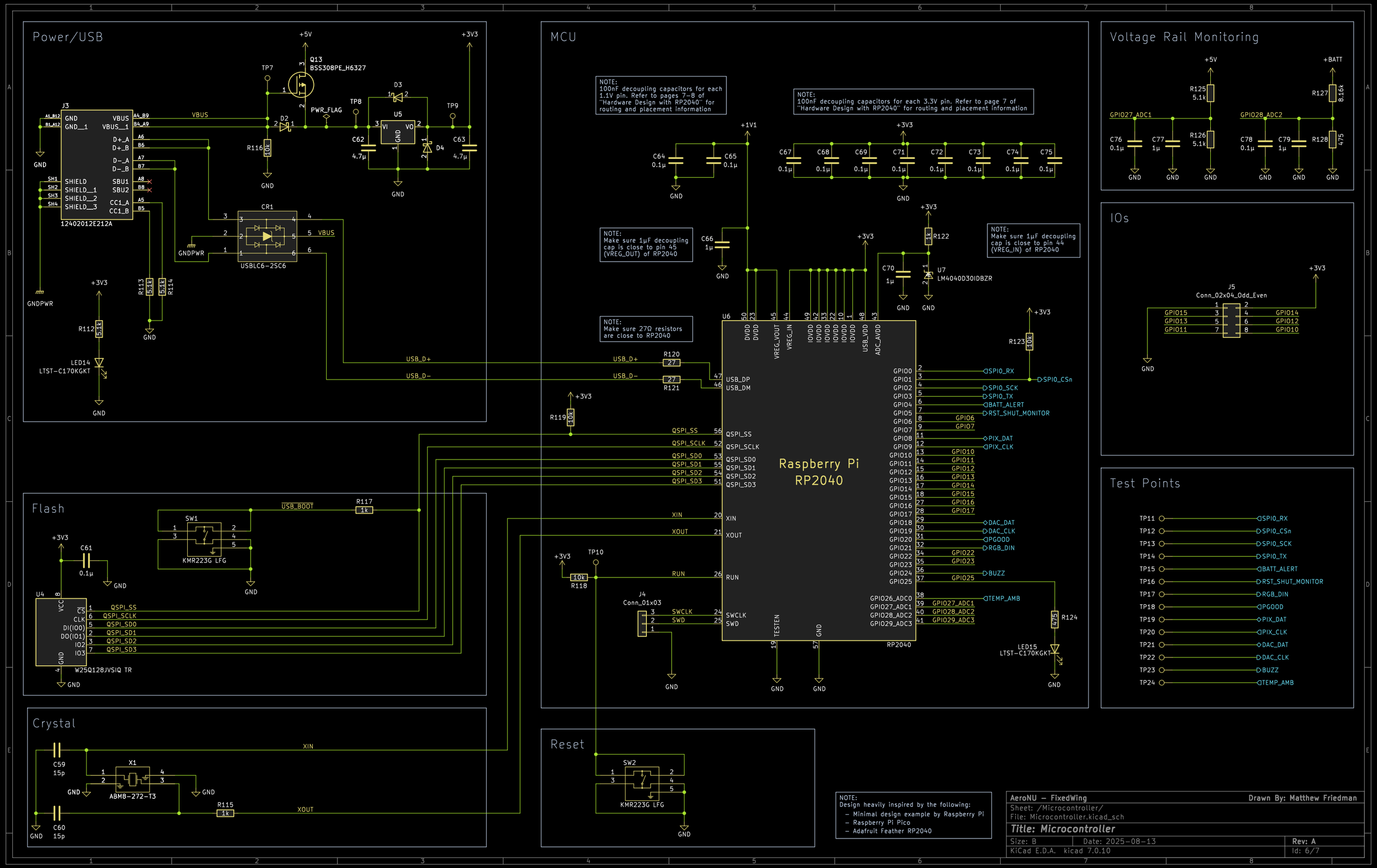
NOTE:
Current sense resistor selected based on p.70 and SRP/SRN connection is based on p.66 of the BQ76972 datasheet.

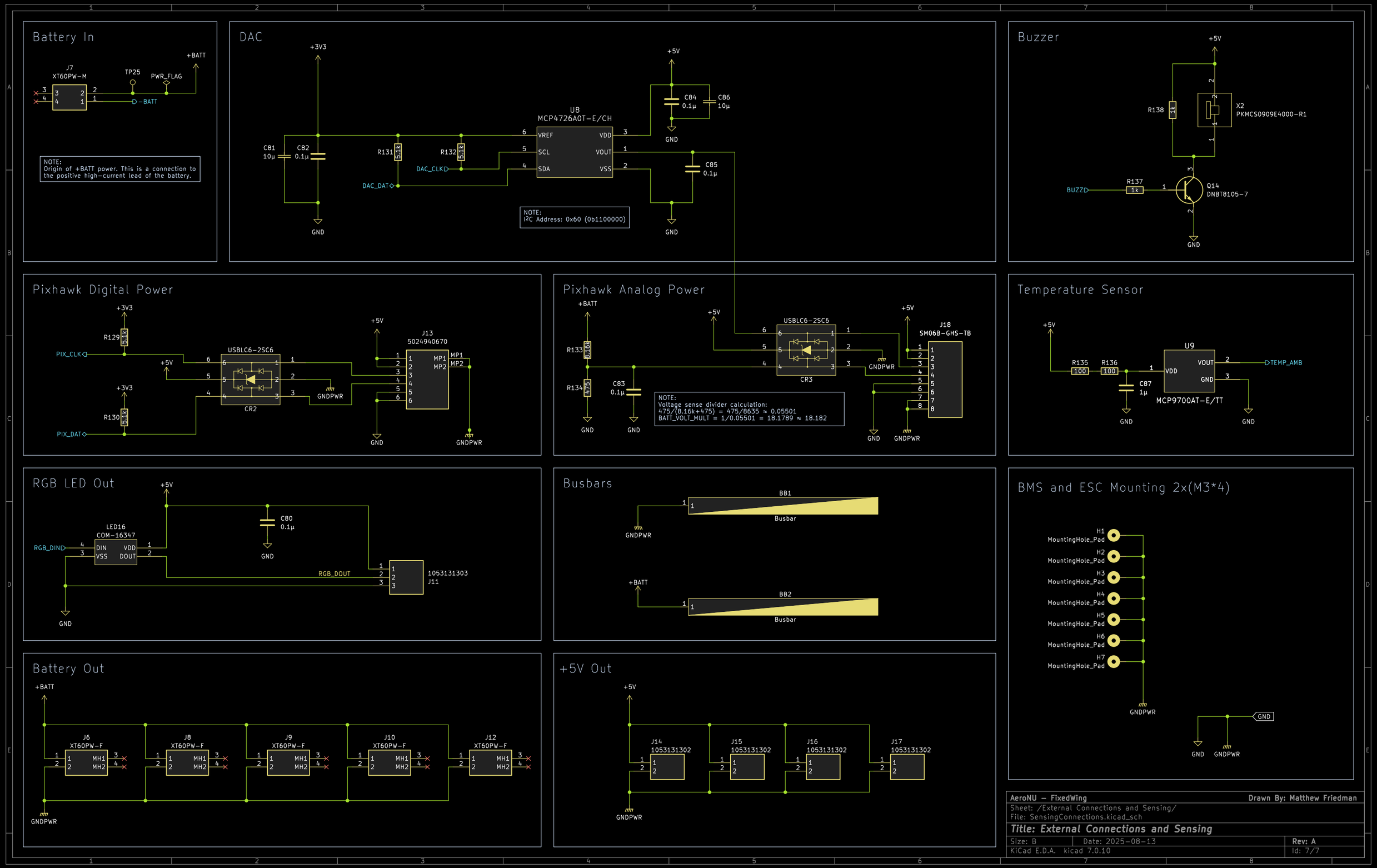
NOTE:
High-current ground connection

AeroNU – FixedWing	Drawn By: Matthew Friedman
Sheet: /Battery Stack Monitor/	
File: BatteryMonitor.kicad_sch	
Title: Battery Stack Monitor	
Size: B	Date: 2025-08-13
KiCad E.D.A. kicad 7.0.10	Rev: A
	Id: 4/7



AeroNU – FixedWing	Drawn By: Matthew Friedman
Sheet: /5V Regulator/	
File: 5VRegulator.kicad_sch	
Title: 5V Regulator	
Size: B	Date: 2025-08-13
KiCad E.D.A. kicad 7.0.10	Id: 5/7
	Rev: A


Power/USB
J3
12402012E212A
VBUS
GND
GNDPWR
Q13
BSS308PE_H6327
+5V
+3V3
TP7
TP8
TP9
PWR_FLAG
D2
D3
D4
U5
C62
4.7µ
C63
4.7µ
R116
10k
CR1
USBLC6-2SC6
R113
5.1k
R114
5.1k
R112
5.1k
LED14
LTST-C170KGKT
MCU
NOTE:
100nF decoupling capacitors for each 1.1V pin. Refer to pages 7-8 of "Hardware Design with RP2040" for routing and placement information
NOTE:
100nF decoupling capacitors for each 3.3V pin. Refer to page 7 of "Hardware Design with RP2040" for routing and placement information
+1V1
C64
0.1µ
C65
0.1µ
C67
0.1µ
C68
0.1µ
C69
0.1µ
C71
0.1µ
C72
0.1µ
C73
0.1µ
C74
0.1µ
C75
0.1µ
NOTE:
Make sure 1µF decoupling cap is close to pin 45 (VREG_OUT) of RP2040
C66
1µ
NOTE:
Make sure 1µF decoupling cap is close to pin 44 (VREG_IN) of RP2040
C70
1µ
R122
1k
U7
LM4040D30IDBZR
NOTE:
Make sure 27Ω resistors are close to RP2040
R120
27
R121
27
R119
10k
R123
10k
U6
Raspberry Pi
RP2040
USB_DP
USB_DM
QSPI_SS
QSPI_SCLK
QSPI_SD0
QSPI_SD1
QSPI_SD2
QSPI_SD3
XIN
XOUT
RUN
SWCLK
SWD
TESTEN
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO15
GPIO16
GPIO17
GPIO18
GPIO19
GPIO20
GPIO21
GPIO22
GPIO23
GPIO24
GPIO25
GPIO26_ADC0
GPIO27_ADC1
GPIO28_ADC2
GPIO29_ADC3
SPI0_RX
SPI0_CSn
SPI0_SCK
SPI0_TX
BATT_ALERT
RST_SHUT_MONITOR
PIX_DAT
PIX_CLK
DAC_DAT
DAC_CLK
PGOOD
RGB_DIN
BUZZ
TEMP_AMB
TP10
R118
10k
J4
Conn_01x03
R124
475
LED15
LTST-C170KGKT
Flash
SW1
KMR223G LFG
USB_BOOT
R117
1k
C61
0.1µ
U4
W25Q128JVSIQ TR
Crystal
C59
15p
X1
ABM8-272-T3
C60
15p
R115
1k
XIN
XOUT
Reset
SW2
KMR223G LFG
Voltage Rail Monitoring
+BATT
R125
5.1k
R126
5.1k
R127
8.16k
R128
475
GPIO27_ADC1
GPIO28_ADC2
C76
0.1µ
C77
1µ
C78
0.1µ
C79
1µ
IOs
J5
Conn_02x04_Odd_Even
GPIO15
GPIO13
GPIO11
GPIO14
GPIO12
GPIO10
Test Points
TP11
TP12
TP13
TP14
TP15
TP16
TP17
TP18
TP19
TP20
TP21
TP22
TP23
TP24
NOTE:
Design heavily inspired by the following:
– Minimal design example by Raspberry Pi
– Raspberry Pi Pico
– Adafruit Feather RP2040
AeroNU – FixedWing
Drawn By: Matthew Friedman
Sheet: /Microcontroller/
File: Microcontroller.kicad_sch
Title: Microcontroller
Size: B
Date: 2025-08-13
Rev: A
KiCad E.D.A. kicad 7.0.10
Id: 6/7



AeroNU – FixedWing	Drawn By: Matthew Friedman
Sheet: /External Connections and Sensing/	
File: SensingConnections.kicad_sch	
Title: External Connections and Sensing	
Size: B	Date: 2025-08-13
KiCad E.D.A. kicad 7.0.10	Id: 7/7
	Rev: A